Homework 2, 4/10/2026

- **(1). (15%)**將你在**Homework 1 (1)**所繳交的圖片使用 **matlab** 來進行 **histogram equalization (**可使用 **matlab** 內建函數或自行撰寫 **histogram equalization)**。並列印出調整前，調整後的圖片，以及對應的 **histogram** 。 並說明你所使用的做法。

- **(2). (15%)**將你在**Homework 1 (1)**所繳交的圖片使用一般影像編輯軟體**(**軟體可自由決定，但以市面或**open source**可公開下載為限**)** 來進行 **histogram equalization** 。並列印出調整前，調整後的圖片，以及對應的 **histogram**。 並說明你所使用的做法。

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❖ (3). 回答下列問題，需有詳細的解答過程。

(70%) Consider the following 5x5 image *A* and a 3x3 Laplacian kernel. The depth of a pixel in these images is 3 bits. The pixel located at the upper left corner is assigned the coordinates (0,0).

Image *A*

2	6	4	7	3
1	5	2	4	0
6	2	**5**	3	7
3	1	1	2	6
2	0	2	1	3

Laplacian kernel

0	-1	0
-1	4	-1
0	-1	0

Image *B*

0	1	6	4	6
5	3	6	5	1
4	3	2	4	7
7	5	3	2	2
2	0	1	1	6

(a) (10%) Calculate the intensity matrix of the histogram-equalized version of Image *A*.

(b) (10%) Calculate the intensity matrix of the histogram-matched version of Image *A* such that the output image has a histogram similar to that of Image *B*.

(c) (10%) What is the intensity of the pixel at the center position (shaded one in the figure) of the Laplacian-sharpened version of Image *A*?

(d) (10%) What is the intensity of the pixel at the center position (shaded one in the figure) of the highboost-filtered version of Image *A*? The blurred image is obtained using a 5x5 box kernel, and the highboost factor is k=3.

(e) (10%) Calculate the full convolution of Image *A* using the kernel provided below.

0.1	0.2	0.3
0.2	1.0	0.2
0.3	0.2	0.1

(f) (10%) Calculate the full correlation of Image *A* using the kernel provided above.

(g) (10%) Using bilinear interpolation, calculate the interpolated intensity at position (2.5, 1.5) of Image *A*.

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作業2之繳交期限: 4/17/2026 11:59AM。

上述照片檔案與說明件之檔名需可辨識是哪題之繳交檔案。將所有檔案**zip**成一個檔案上傳到**E3**。上傳檔案之檔名需包含同學之學號與姓名，作業次。

如**: ip-**王曉明**-112222333444-HW2.zip**